

View Reviews

Paper ID

35

Paper Title

Agentic Architecture for Data-Centric AutoML

Track Name

Research Papers

Reviewer #1

Questions**1. Brief summary (1 paragraph) of problem addressed and contributions of the paper.**

The paper presents a multi-agent architecture to automatically generate ML pipelines from a user specification. They propose a data-centric approach where not only the model and its hyperparameters are optimized, but also the data preparation pipeline.

2. Overall Recommendation

Weak Accept

3. Strong Points (at least 3)

- The approach is novel
- Code is openly provided
- Experimental results seem promising

4. Weak Points (at least 3)

- Explanations are a bit poor and confusing (specially in the experimental results)
- Presentation should be improved
- Lacks some formality and precision in the explanations and claims

5. Overall evaluation

The idea is promising and the work interesting, with an implementation publicly available. However, there are clear indications of the paper being written and submitted in a rush (eg, using more space than allowed, while some figures and tables could have been easily reduced in a minute to compensate that).

6. Reviewer's Confidence

Moderate Confidence

9. Required revisions?

Yes

10. Specific issues to be addressed in a revision.

- Figure 4 size should be reduced
- Font size in table 1 should be reduced
- Caption of table 1 should be underneath (like in the others).
- Figure 1 should have a legend with the meaning of the arrows.
- Machine Learning should be shortened as ML everywhere.
- Listing 1 is fine, but a grammar should be provided
- Being a multi-agent system is only mentioned in the introduction, but later it is not that explicit (eg, Figure 3 is confusing)
- At some point seems that only classification problems can be solved, but then the experiments show also regressions.
- Experiments should be provided on the effect of the budget (how the number of generated pipelines helps? How the number of executed pipelines help?)
- Where are "supplementary materials"? does this refer to the GitHub repository? In which file/folder?
- Dataset characteristics provided in table 1 are interesting, but they not necessarily reflect the complexity of the problem.
- The meaning and interpretation of the different columns in tables 2 and 3 is not clearly explained.
- More details should be provided on the separate cost and impact of each one of the agents.
- The claim about the balanced distribution between LLM inference and ML training is not clearly explained. Where exactly does it come from?

Reviewer #2

Questions

1. Brief summary (1 paragraph) of problem addressed and contributions of the paper.

This paper presents AGE-ML, an agentic framework designed to align Automated Machine Learning (AutoML) with the principles of data-centric AI and MLOps.

Traditional AutoML systems often act as "black boxes" focused solely on model selection and hyper-parameter tuning, which limits their controllability and transparency. AGE-ML introduces a human-readable specification language (YAML-like) that allows data scientists to encode domain knowledge and constraints. These specifications are used by a Multi-Agent System (MAS) that combines symbolic reasoning (via a Z3 CSP solver) to find admissible pipelines and LLMs to generate, validate, and explain executable Python code. The framework integrates with MLflow for hierarchical

experiment tracking to ensure traceability and reproducibility.

2. Overall Recommendation

Accept

3. Strong Points (at least 3)

Unlike standard LLM-based code generators, AGE-ML uses a Constraint Satisfaction Problem (CSP) solver to guarantee that every generated pipeline strictly adheres to the expert's structural constraints and domain rules.

The framework prioritizes traceability and reproducibility by using a hierarchical data catalog (MLflow) that logs every script, hyper-parameter assignment, and natural-language explanation.

The experiments demonstrate that the framework is effective even when using smaller, cost-efficient LLMs (Gemini 3.1 Flash Lite and Gemini 2.5 Flash), matching or surpassing state-of-the-art baselines like HAMLET on several benchmark datasets.

The system generates structured, human-readable descriptions of the design rationale for each pipeline, addressing the "opacity" issue in traditional AutoML.

4. Weak Points (at least 3)

The effectiveness of the search space exploration is heavily dependent on the LLM's ability to generate valid code; in some tests (e.g., HAR and MNIST 784), the system failed to explore the full budget because the models could not consistently produce valid pipelines.

In multiple datasets (e.g., Adult and California Housing), LLM inference accounted for more than 70% of the total execution time, indicating potential inefficiencies in prompt design or the agentic loop.

While the framework allows for expertise injection at the start, it currently lacks the capability for data scientists to directly edit or selectively re-execute specific components of a pipeline once the automated process has begun.

5. Overall evaluation

AGE-ML is a timely and well-engineered contribution that successfully bridges the gap between automated optimization and human expertise. By shifting the focus from "black-box" model tuning to a specification-driven approach, it provides the controllability necessary for professional MLOps environments. The empirical results across classification and regression tasks are strong, and the use of symbolic constraints to bound LLM outputs is a robust architectural choice.

6. Reviewer's Confidence

Moderate Confidence

9. Required revisions?

Yes

10. Specific issues to be addressed in a revision.

The authors should discuss strategies for reducing the high inference latency and redundant conversational context mentioned in the limitations.

Elaborate on how future versions might move beyond symbolic enumeration toward more efficient heuristic-based search strategies to improve exploration diversity.

Provide more detail on the feasibility of the proposed "direct editing" feature, as this is a critical requirement for true data-centric AI.

Reviewer #3

Questions**1. Brief summary (1 paragraph) of problem addressed and contributions of the paper.**

The authors present an agentic architecture to automatically generate pipelines based on experts inputs (via a YAML-like description of their requirements). The generated pipelines are expected to be more transparent and more reproducible because they are generated based on a human expert and their are validated.

2. Overall Recommendation

Weak Accept

3. Strong Points (at least 3)

S1. The paper is well written and clear.

S2. The use of both LLM judgement but also rules in the form of regular expressions to check pipeline generation

4. Weak Points (at least 3)

W1. LLMs are known to be not deterministic. How do you manage that? (especially in terms of reproducibility)

W2. The reproducibility aspect is not enough emphasized in the paper. It would be nice to have feedback / an experiment on how the pipelines are more transparent and/or more reproducible compared to existing systems.

5. Overall evaluation

Overall, the paper is clear and sound. The usage of LLM is interesting despite the

accumulation of agents and LLMs introduces uncertainty and non-determinism at many steps.

Few more remarks:

- how is the compact description of the dataset structure obtained? Is it metadata / generated by an agent / ...?
- a small description of each dataset would be helpful
- use bold font for numbers surpassing the baselines would be helpful to read more easily the table

Typos:

- p13 : space after the parenthesis "(Wilt..."
- same for "(California..."

6. Reviewer's Confidence

Moderate Confidence

9. Required revisions?

Yes

10. Specific issues to be addressed in a revision.

Please address W1 and W2.

Reviewer #4

Questions

1. Brief summary (1 paragraph) of problem addressed and contributions of the paper.

This paper proposes AGE-ML (Agentic Architecture for Data-Centric AutoML), a novel AutoML architecture driven by AI agents and Large Language Models (LLMs), designed to be more transparent, explainable, and data-centric.

2. Overall Recommendation

Weak Reject

3. Strong Points (at least 3)

- The paper is well written and very clear. The proposed architecture is well defined, and its different components are thoroughly described.
- The paper distinguishes itself from the state of the art by adopting a data-centric rather than a model-centric perspective. This is an important contribution, as data quality and preprocessing have a significant impact on decision-making and machine learning performance.

- The experimental evaluation is interesting and relies on a diverse set of benchmark datasets.
- The authors make the source code publicly available through a GitHub repository, which supports reproducibility.

4. Weak Points (at least 3)

- The formalization of expert knowledge (in this case, data scientists' knowledge) deserves further elaboration. The paper assumes the existence of a single specification expressing the knowledge of a data scientist, but it does not discuss situations where multiple experts contribute to defining constraints. In particular, mechanisms for conflict resolution, prioritization, or the integration of contradictory knowledge are not addressed.
- The evaluation relies primarily on traditional predictive performance metrics. Given the data-centric nature of the proposed approach, I would have expected a more data-oriented evaluation, including metrics related to data quality, improvements to the dataset, error detection, data cleaning effectiveness, or bias reduction. This is somewhat paradoxical, as the paper advocates a data-centric perspective, yet this aspect is largely absent from the evaluation, where datasets are already provided as inputs. As a result, it remains unclear how AGE-ML concretely contributes to improving data quality.
- The paper claims to improve explainability; however, the mechanisms presented are mainly related to traceability and documentation of the pipeline generation process. The explainability of the resulting models themselves is neither addressed nor experimentally evaluated. Consequently, the use of the term "explainable" appears insufficiently justified.

5. Overall evaluation

Overall, the paper presents an interesting approach that may provide valuable contributions to the community. However, several aspects deserve further investigation and development, particularly regarding the impact on data quality, the evaluation protocol, and the integration and management of expert knowledge.

6. Reviewer's Confidence

Moderate Confidence

9. Required revisions?

No